Program : **Diploma in Cloud Computing and Big Data**	
Course Code : **3275**	Course Title: **Database Lab**
Semester : **3**	Credits: **1.5**
Course Category: **Program Core**	
Periods per week: **3 (L:0 T:0 P:3)**	Periods per semester: **45**

Course Objectives:

- Impart the design and management of relational databases using SQL.
- Familiarize with the design and management of NOSQL databases.

Course Prerequisites:

Topic	Course code	Course name	Semester
Basic programming concepts	1008	Introduction to IT systems Lab	I
Programming concepts	2131	Problem Solving & Programming	II
	2139	Problem Solving & Programming Lab	II

Course Outcomes:

On completion of the course, the student will be able to:

COn	Description	Duration (Hours)	Cognitive Level
CO1	Demonstrate the use of DDL and DML commands in database	10	Applying
CO2	Illustrate SQL queries involving aggregate functions, join operations, views and nested queries	11	Applying
CO3	Construct SQL programs using functions, procedures, and trigger	11	Applying

CO4	Implement document based NOSQL databases using MongoDB	10	Applying
	Lab Exam	3	

CO – PO Mapping

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3	3	3	2	-	-	-
CO2	3	3	3	2	-	-	-
CO3	3	3	3	2	-	-	-
CO4	3	3	3	2	-	-	-

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcomes	Name of the Experiment	Duration (Hours)	Cognitive Level
CO1	**Demonstrate the use of DDL and DML commands in database**		
M1.01	Demonstrate DDL commands-CREATE, ALTER, DROP	2	Applying
M1.02	Apply integrity constraints - PRIMARY KEY, NOT NULL, DEFAULT, CHECK, UNIQUE, FOREIGN KEY	2	Applying
M1.03	Experiment DML commands - INSERT, SELECT, UPDATE and DELETE to manipulate databases	3	Applying
M1.04	Use WHERE, DISTINCT, BETWEEN, ORDER BY, IN, LIKE, set operations in SELECT statements for data retrieval from the databases	3	Applying
CO2	**Illustrate SQL queries involving aggregate functions, join operations, views and nested queries**		

M2.01	Implement aggregate functions -SUM, AVG, COUNT, MIN, MAX - GROUP BY and HAVING clause	3	Applying
M2.02	Construct nested queries	3	Applying
M2.03	Demonstrate SQL queries for inner and outer joins.	3	Applying
M2.04	Illustrate various views of the table	2	Applying
	Lab Exam – I	1.5	
CO3	**Construct SQL programs using functions, procedures, and trigger**		
M3.01	Demonstrate the stored procedures for simple tasks	3	Applying
M3.02	Apply stored functions for database manipulations	4	Applying
M3.03	Illustrate SQL programs using triggers	4	Applying
CO4	**Implement document based NOSQL databases using MongoDB**		
M4.01	Demonstrate the Installation and configuration of MongoDB	2	Applying
M4.02	Experiment the functions of Mongoengine library - connect, Document, ListField, EmbeddedDocument	3	Applying
M4.03	Illustrate the use of insert, save, print, update and delete functions in MongoDB	5	Applying
	Lab Exam – II	1.5	

Text / Reference

T/R	Book Title/Author
T1	Elmasri, Navathe, Fundamentals of Database Systems, 7th Ed., Pearson
R1	Sliberschatz A., H. F. Korth and S. Sudarshan, Database System Concepts, 6th ed, McGraw Hill, 2011.
R2	ITL Education Solutions ltd, Introduction to Database Systems, Pearson,2011

Online Resources

Sl No	Website Link
1	https://www.w3schools.com/sql
2	https://www.tutorialspoint.com/sql
3	https://www.javatpoint.com/sql
4	https://www.guru99.com/sql
5.	https://webcms3.cse.unsw.edu.au/COMP9321/18s1/resources/16130
6.	https://docs.mongodb.com/manual/tutorial/getting-started/